

Thuan Doan

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SUMMARY

AI Engineer with 10+ years of software engineering experience, including 5 years in AI, building machine learning, LLM, retrieval systems, and agentic workflows for production. Currently interested in physical AI and world models.

EXPERIENCE

Spartan

Nov 2024 – Present

AI Engineer Lead

Ho Chi Minh, Vietnam

- Supported solving AI problems across projects within the company.
- Conducted technical sharing, mentored junior engineers, and supported AI hiring and interviews.

TriangleHealth | *Apache Airflow, TxGemma, GraphRAG, Neo4j, Google ADK, MCP, vLLM*

- Built Airflow-based data pipelines processing 10M+ biomedical entities and documents, enabling large-scale knowledge graph construction in Neo4j.
- Designed graph enhanced retrieval (hybrid) + reranking (TourRank) improving answer accuracy by +15-20% on internal medical retrieval benchmarks.
- Developed multi-agent medical research system (Google ADK + TxGemma/BioMNI) to minimize hallucinations and deliver evidence-grounded, verifiable outputs.
- Optimized model deployment using vLLM. Implemented tracing and observability with Langfuse and Datadog.

LayerProff | *OpenAI API, Nano Banana, LangGraph, FAISS, Qdrant, python-pptx, FastAPI*

- Built a multi-agent AI system that transforms documents into professional presentation slides (PPTX) using specialized agents for planning, writing, designing, and refining.
- Fine-tuned open-source image models (FLUX.1-dev, SDXL) with LoRA for presentation-ready visual generation.

QAMobileAgent | *GUI-Owl-7B, Appium, uiautomator2, Hugging Face Transformers, FastAPI, LoRA*

- Designed a hybrid GUI testing agent for banking apps combining a multimodal vision-language model (GUI-Owl-7B) with deterministic Appium execution and rule-based safety fallbacks.
- Built structured UI state representation using screenshots, accessibility trees, and OCR. Designed fine-tuning roadmap with SFT, preference optimization for domain adaptation.

HCLTech

Jun 2023 – Oct 2024

AI Engineer/Researcher

Ho Chi Minh, Vietnam

FinGPT | *Azure OpenAI, LangChain, LangGraph, FastAPI, Langfuse, Promptfoo, DeepEval*

- Built a multi-agent conversational AI system for retail banking, integrating planner, retrieval, banking API, and compliance agents, supporting 1K+ daily user queries.
- Designed and implemented a production-grade RAG pipeline (hybrid search + re-ranking), improving response accuracy by 20–25% on internal evaluation benchmarks.
- Developed compliance and safety layer (PII detection/redaction, audit logging, role-based retrieval), reducing risk of sensitive data exposure and ensuring 100% audit traceability.

EnsayoAI | *Llama3, LangGraph, RAG, Langfuse, Promptfoo, DeepEval, pgvector, FastAPI, AWS SageMaker*

- Developed a generative AI testing platform leveraging Llama 3 and Mistral, automating test case generation and reducing manual QA effort by 40–60%.
- Applied domain-adaptive fine-tuning (LoRA, RLHF), combining with heuristic rules to improve relevance and correctness of generated test cases by 15–20% based on evaluation metrics.
- Designed and deployed LLM evaluation pipelines (DeepEval, Promptfoo) to benchmark model performance and ensure reliability across test scenarios.

Infinigr

Aug 2018 – Apr 2023

Machine Learning Engineer

Korea/Vietnam

GruBot | *Python, wav2vec, Kaldi, RASA, BERT, LSTM, Kubernetes, MLflow, Prometheus, Grafana*

- Developed and fine-tuned Korean and Vietnamese speech recognition models using Kaldi and wav2vec.
- Built RASA-based conversational AI chatbot with NLU pipeline and dialogue management, handling multi-turn conversations with seamless human agent hand-off.
- Optimized BERT-based models for Named Entity Recognition (NER) and Sentiment Analysis.

ViettelFDS | *Python, Java, Spring Boot, Apache Kafka, Apache Flink, PySpark, XGBoost, LightGBM, SHAP, VertexAI*

- Developed high-performance real-time fraud detection system combining rule-based systems with ML models (Random Forest, XGBoost, LightGBM) to reduce false positives and detect evolving fraud patterns.
- Built real-time data pipeline using Apache Kafka and Flink for processing payment transactions with threshold-based decision-making.

AirPatrol | *U-Net, Mask R-CNN, TensorFlow Lite, ONNX, TensorRT, Jetson Nano*

- Developed and deployed deep learning models (U-Net, Mask R-CNN, Xception) for crack detection and wildfire classification, optimized with TensorFlow Lite for real-time inference on NVIDIA Jetson Nano.

Earlier Experience

2008 – 2018

Global Outsource Solution (Technical Lead), Global CyberSoft, Foria (Software Engineer)

- Led development of scalable web and enterprise systems across e-commerce, CMS, and data platforms.
- Built backend services and system architectures; established CI/CD pipelines and engineering best practices.

EDUCATION

University of Information Technology

Ho Chi Minh, Vietnam

Master's Degree in Computer Science

2020 – 2023

- Research assistant at MMLab. Topics: NLP, LLMs, VLM, SLAM, 3D vision, deep generative models.

Japan Advanced Institute of Science and Technology (JAIST)

Ishikawa, Japan

Research Assistant under supervision of Prof. Kazuhiro Ogata

Jan-Mar 2020

- Research Topic: Theorem proving and state machine with CafeOBJ.

Kookmin University

Seoul, Korea

Research Assistant under supervision of Prof. Young Kim.

2011 – 2012

- Research Topic: Security-Enhanced Linux (SELinux) for web servers.

University of Science

Ho Chi Minh, Vietnam

Bachelor's Degree in Computer Science

2004 – 2008

TECHNICAL SKILLS

Programming: Claude Code, Python, JavaScript, Java, PHP

AI Frameworks & Tools: TensorFlow, PyTorch, Scikit-learn, Hugging Face Transformer, Pandas, Numpy, Matplotlib, OpenCV, NLTK, MLflow, Datadog, Sentry, Prometheus, Grafana, SageMaker, VertexAI.

LLM Tools: LangChain, Google ADK, LlamaIndex, Langfuse, LiteLLM, vLLM, Ollama, Unsloth, DeepEval.

App Frameworks: FastAPI, Flask, Spring, Laravel, React, NodeJS, VueJS

Data & Cloud: MySQL, PostgreSQL, MongoDB, Redis, Neo4j, Airflow, Kafka, Flink, PySpark, ChromaDB, AWS, GCP.

HOBBY PROJECTS

ChefBuddy | *NVIDIA GR00T N1.5, LeRobot, SO-100/SO-101, LoRA, TensorRT*

- Built a kitchen robot using NVIDIA's GR00T N1.5 vision-language-action model with SO-100 arm for language-conditioned pick-and-place tasks such as sandwich assembly.
- Implemented full pipeline from teleoperation data recording to LoRA fine-tuning and real-time deployment.

HomeExplorer | *ROS, GMapping, Hector SLAM, YOLOv5, TensorRT, Jetson Nano*

- Built an edge-based SLAM and semantic mapping system combining 2D LiDAR localization with camera-based object detection on NVIDIA Jetson Nano.

ADDITIONAL

Certificates

- Tensorflow Developer - [Certificate](#)
- Coursera specializations about Math, Machine Learning, Deep Learning, Vision, NLP, Data Science - [Certificates](#)
- IELTS 6.5 - [Certificate](#)

Publication:

- Efficient Finetuning Large Language Models For Vietnamese Chatbot [MAPR](#) 2023